

Programming Assignment

Diffuse and Stochastic Processes

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Declaration of AI used I used Copilot in VS Code to help formatting in plotting, and find the right syntax for package functions faster.

Code at GitHub The code and images used here can also be found at my GitHub repository:

<https://github.com/resch-peter/Diffuse-and-Stochastic-Processes>

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1 Langevin simulation for a biased double well potential

1.1 Motivation for model

In my simulation I am going to make a biased brownian motion in a double well potential $U(x)$. This could simulate cyclists riding along a road, which has some curvature. In the middle it is the highest and falling off on both sides - so its not easy to always ride on the highest point. No one wants to cycle next to the road in the grass, so this is also accounted for with the repellent effect of the potential far off. There is also a constant term, that could be seen as wind coming constantly from one side and pushing the cyclists towards one side.

1.2 Equations and variable definitions

The Langevin equation, which describes the position on the curvature of the street over time is

$$\frac{dX}{dt} = X - \frac{1}{\eta} \frac{dU(X)}{dx} + W + \Delta R,$$

where $\langle \Delta R^2 \rangle = 2D\Delta t$ and $\langle R \rangle = 0$. The potential selected is

$$U(x) = \frac{1}{2}(x^4 a - x^2 b),$$

which can be seen in Fig.1 and therefore

$$\frac{dU}{dx} = 2x^3 a - xb.$$

η is the viscosity and could be interpreted in this model as the reactiveness of steering of the cyclist. ΔR are the random deviations of steering during cycling.

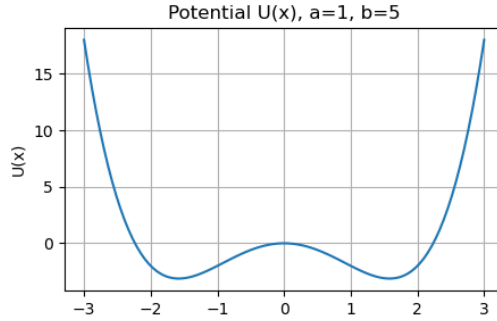


Figure 1: Position dependent double well potential $U(x)$ for the parameters $a = 1, b = 5$, which gives a potential force in this system.

Parameter	Value	Unit
w	1.0	m/min
a	1.0	1/m ²
b	5.0	1
η	2.0	1/min
Diffusion coefficient	2.0	1/min
Initial position	-2.0	m

Table 1: Parameter values

1.3 Predictions

My prediction without wind is, that over time the cyclists will rather be more in the valleys of the potential, sometimes switching sides, depending on how high the potential in the middle is or

how strong the random term (the diffusion coefficient) is respectively. This means, that the mean of the ensembles should be zero and the variance reaches a steady state, as the potential gives boundaries to the diffusion.

With the wind though, in the long run the trajectories get pushed more to one side, through which the mean should also be pushed to one side. For me it is hard to make predictions on the variance on a general case in this model, as it could be lower or higher than in the case without wind, as $Var = \langle (X - \langle X \rangle)^2 \rangle$ and therefore the deviation far away from the mean counts more, but might also be rare. So, the variance is depending on the parameter values of the potential and the wind bias, I suspect.

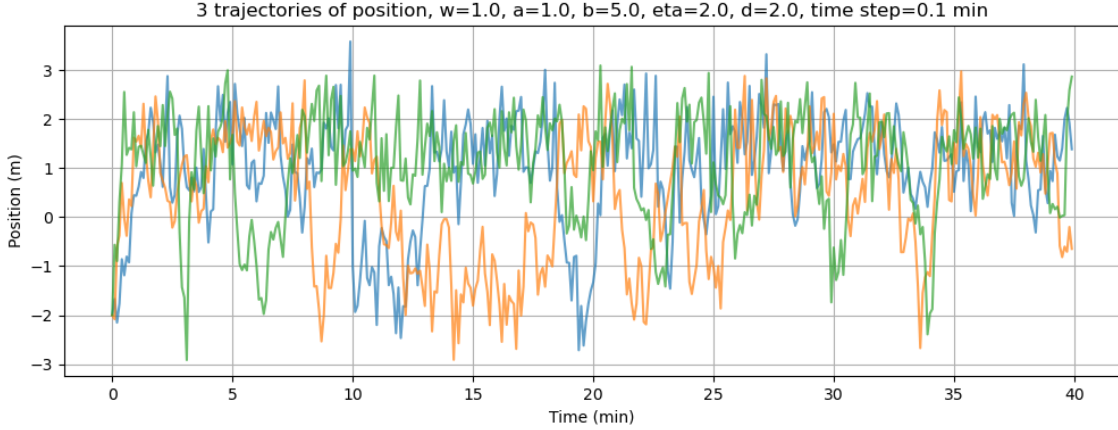


Figure 2: Trajectories of 3 simulations, where one can see, that the position of the potential valleys are preferred over time.

1.4 Summary of results and comparison to predictions

The Langevin model for this system and for this set of parameters (Tab.1) has a drift towards one side of the double well potential, which was plotted for 3 trajectories and displayed in Fig.2. The ensemble behavior over time can better be seen in the histogram (Fig.3) and in the mean position (Fig.4).

The mean position over time was landing in one valley of the potential - although not in the deepest spot of the potential, depending on the wind term, a little more to the outer side of the valley, just as predicted. The deviation in the histogram between the different time step simulations can be explained by the introduced error by this kind of numerical integration, which scales proportional with the time step size Δt .

For the variance, there is a clear jump visible in the beginning as the position starts at a position far off from the steady state mean. After this initial phase the system tumbles around its ensemble mean, through which also the variance stabilizes.

I came hereby to the insight, that the variance is always smaller in the long run, than in the case without wind. This is because $Var = \langle (X - \langle X \rangle)^2 \rangle$ has its maximum, where $\langle X \rangle = 0$.

One more insight was, that the constant wind term could also be put in the potential term, through which the potential gets distorted to one side and the same model emerges.

The biggest insight I had during experimenting was, that I could not put the wind bias in the noise term, like in this piece of code:

```
def brownian_move(diffusion_coef, time_step, bias):
    return np.random.normal(bias, np.sqrt(2 * diffusion_coef * time_step))

def next_position(...):
    return current_pos + brownian_move(...) + pot_f(...) * t_step
```

This produced a time step dependent behavior, as seen in Fig.5 and costed me some time to figure out, where the error was.

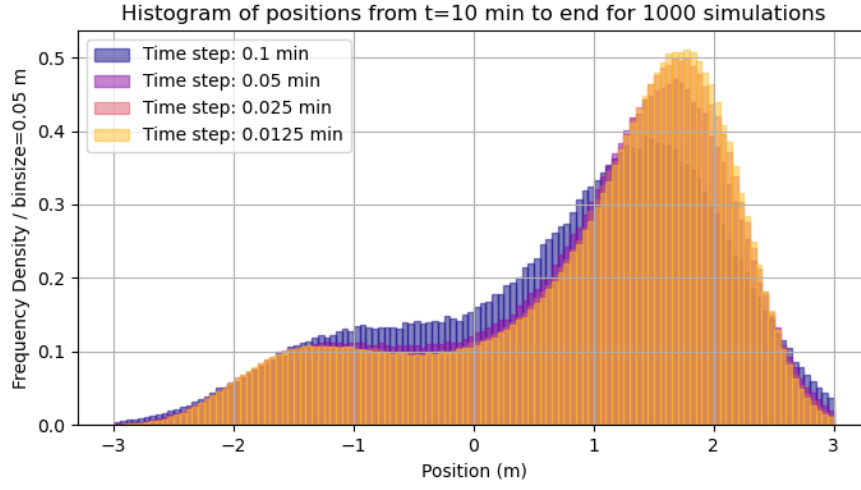


Figure 3: Histogram of positions for 1000 simulations for different time step sizes.

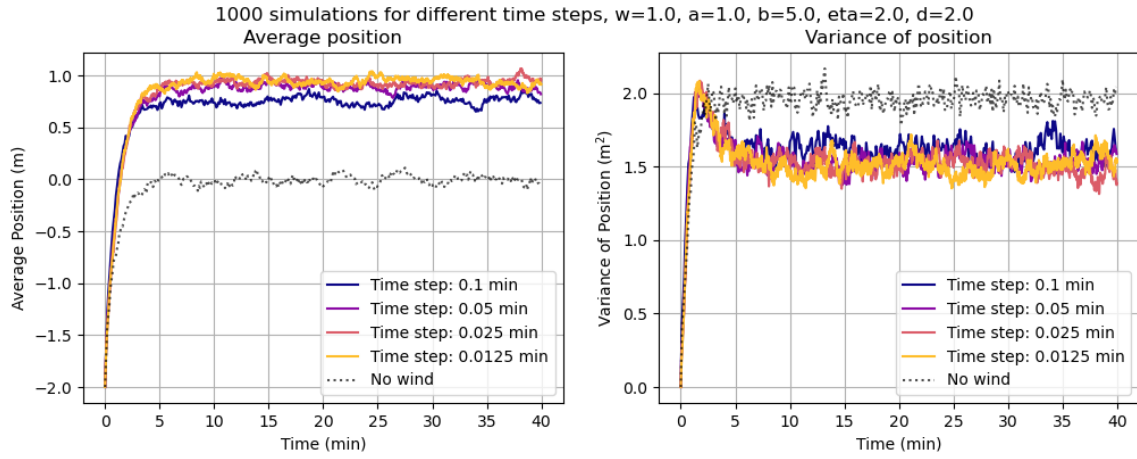


Figure 4: Mean of position and variance over 1000 simulations over different timestep sizes. In dotted black, one can also see the case without wind, calculated with timestep size 0.1 min.

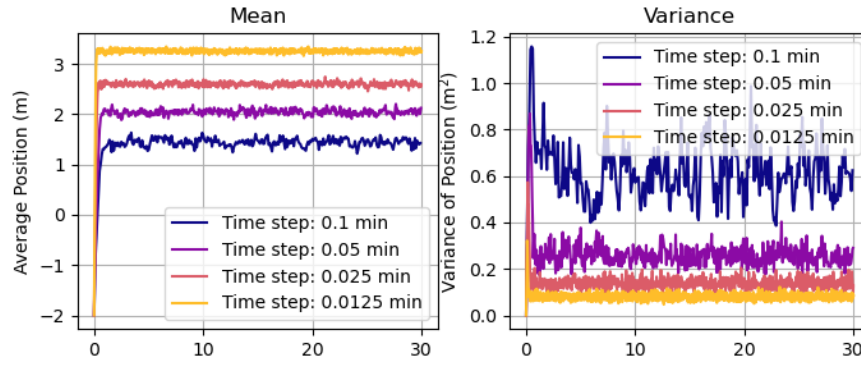


Figure 5: Mean of position and variance over 100 simulations over different timestep sizes with a wrong program code, resulting in a time step size dependent behavior.

2 Gillespie model of a kombucha fermentation metabolism

2.1 Motivation for model

When fermenting food, one type of bacteria group is often found - acid bacteria. In a rough explanation, they essentially metabolize sugars into acids and other metabolites, which we will not describe further for this simplified model. For fermenting purposes, this is often wanted in terms of longer preservation of food by lowering the pH level or also for flavors (e.g. yogurt, salami, sauerkraut, kombucha).

One of the first steps in fermentation is to add starter culture, to introduce a major amount of the wanted bacteria to the substrate, so that they can take the lead in civilizing before other microorganism can do this. You could also do this without, which is then often called wild fermentation. The lowering of pH levels is a great way to preserve foods, as each microorganism has their own preferred milieu (pH-value, temperature, substrate, air supply) and the acid bacteria group is one, which like it sour and also produce acids themselves. I came up with an idea to model a metabolite process of kombucha - a tea based drink, which ferments sugar over time, where acetic acid bacteria play a major role.

2.2 reaction scheme, propensity functions and parameter values

To keep this model as simple, but still functional as possible, I selected three processes, which are relevant and three amounts of populations, which interact with each other. There is the amount of Sugar given as S , the amount of H^+ ions in the liquid as H and the population of active bacteria N . As the dynamics in this model depend on populations in a certain volume, these sizes are defined in units per liter.

The population of bacteria can grow by eating sugar, the rate depending on the concentration N and S in the volume, where λ is the coefficient of growth:

$$g(N, S, \lambda) = NS\lambda \quad , \quad N + S \rightarrow (N + 1) + (S - 1)$$

The population can also become really slow in reproduction and metabolizing (deactivated) or dies, which is dependent on the deviation from the neutral pH level (pH=7) of the liquid and should resemble a exponential decay:

$$d(H, N, \gamma) = (7 - pH(H))\gamma N \quad , \quad N \rightarrow (N - 1)$$

The bacteria can also metabolize one sugar molecule to $\alpha * H^+$ ions - depending on the sugar molecule - without growing, which in an simple assumption has a multiplicative rate to the growth process by μ , as it should be dependent on the same parameters:

$$m(N, S, \lambda, \mu) = \mu g(N, S, \lambda) \quad , \quad S + H \rightarrow \alpha * (H + 1) + (S - 1)$$

I will introduce a upscaling for simulating the S and H populations by 10^{10} , as the pH is connected to mol and these would be quite small quantities and I would like to make discrete steps for the rates. I also thought about expressing the populations with their "real" amount (with avogadro's number), but this would be then too big for me, so I settled on this chosen scaling factor.

Parameter	Value	Unit
λ	10^{-8}	s^{-1}
γ	10^{-4}	s^{-1}
α	1.0	1
μ	2.0	1

Table 2: Parameter values

2.3 Expected mean behavior

My expectation of the behavior is, that after initial exponential growth of the bacteria population and H^+ , the growth will flatten and stabilize due to a higher pH level. The pH should still be raising, until all sugar is consumed. Thereafter, the pH should stay the same and the bacteria population should continuously go down, with exponential decay.

2.4 Fluctuation prediction

In the period of exponential growth, I suspect large deviation for each trajectory from the mean and therefore in the variance. As the system stabilizes due to the higher pH levels, I suspect that the variance could be small, relative to the mean. For the exponential decay of the population after the sugar is consumed, again large variance is suspected. My understanding of the inherence of these dynamics is due to the battling exponential growth and decay process structures of the model.

2.5 Selected initial condition

Initial condition	Value	Unit
N	100	10^{-10} mol/liter
S (0.01 g saccharose/liter)	292397	10^{-10} mol/liter
H (pH=7)	1000	10^{-10} mol/liter
Total time	10000	s

Table 3: Selected initial condition

In Fig.6 one can see the population plot over time for these initial conditions for 3 simulations.

2.6 Calculating system statistics

In Fig.7 one can see the mean, variance and variation coefficient for the 3 populations over time for the afore mentioned initial conditions.

2.7 Summary of results and comparison to predictions

This Gillespie model of a simplified kombucha fermentation metabolism with this specific set of initial parameters shows, that as long there is sugar S , the H molecules are increasing and no more change in H and S , when all the sugar is consumed. This was already expected in the predictions. In the beginning, starting low, the bacteria population N sees exponential growth, while there is sugar and still not a too acidic milieu. When the acidity increases, the growth of N dampens and as the system with these start parameters ran out of sugar, the bacteria dynamic goes from growing into degrading. In all 3 populations, the variance peaked in the growing phase of N or more specific, it seems to peak at the turning point of the slope (around $t=2500$). Differing from the predictions, I would say now, that not only is the fluctuation coming from the exponential growth and decay, but foremost on the time difference of this turning point in the different simulations and thereby the temporal change of dynamics.

After having consumed all the sugar, the bacteria experiences an exponential decay on another timescale as growth. This and also the exponential growth cohere with the predictions. In the predictions, I postulated also a stabilizing phase, where growth and degradation keep balance. This did not manifest with these initial conditions, because I suspect, that the sugar ran out, before it could get there.

Anyways, after seeing the these plots, I would now predict, that there is not a real stabile phase, as the pH still changes and therefore also the degradation rate. I guess therefore, that N would peak and get more to a almost linear slow decrease, depending on the interplay of alpha, mu, lambda and gamma until the sugar is consumed.

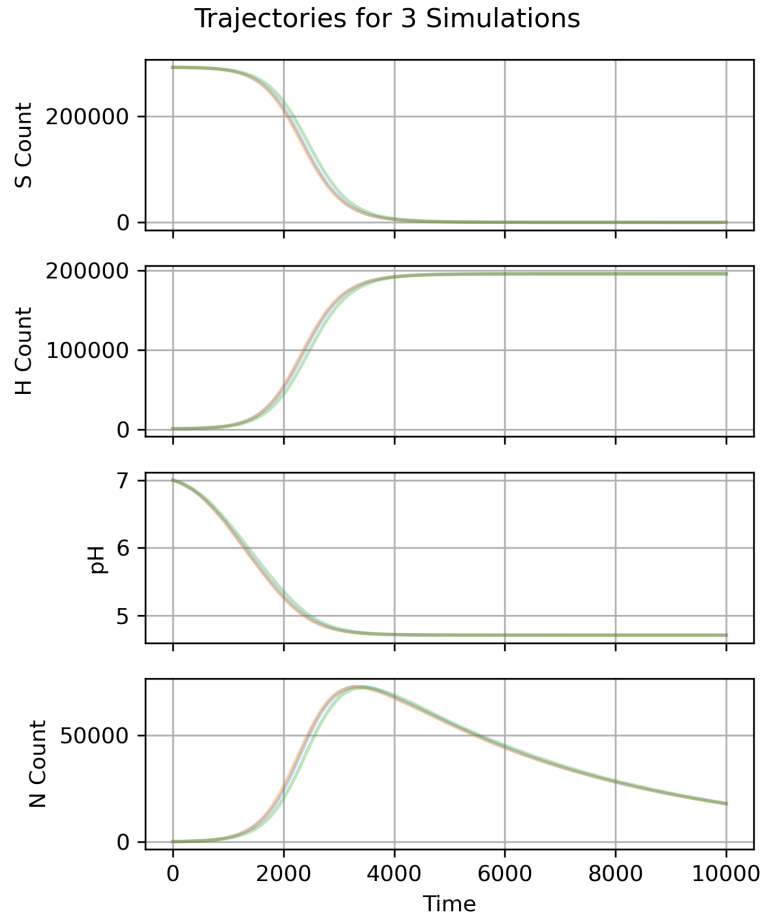


Figure 6: Trajectories of 3 simulations of the 3 populations over time, including pH levels.

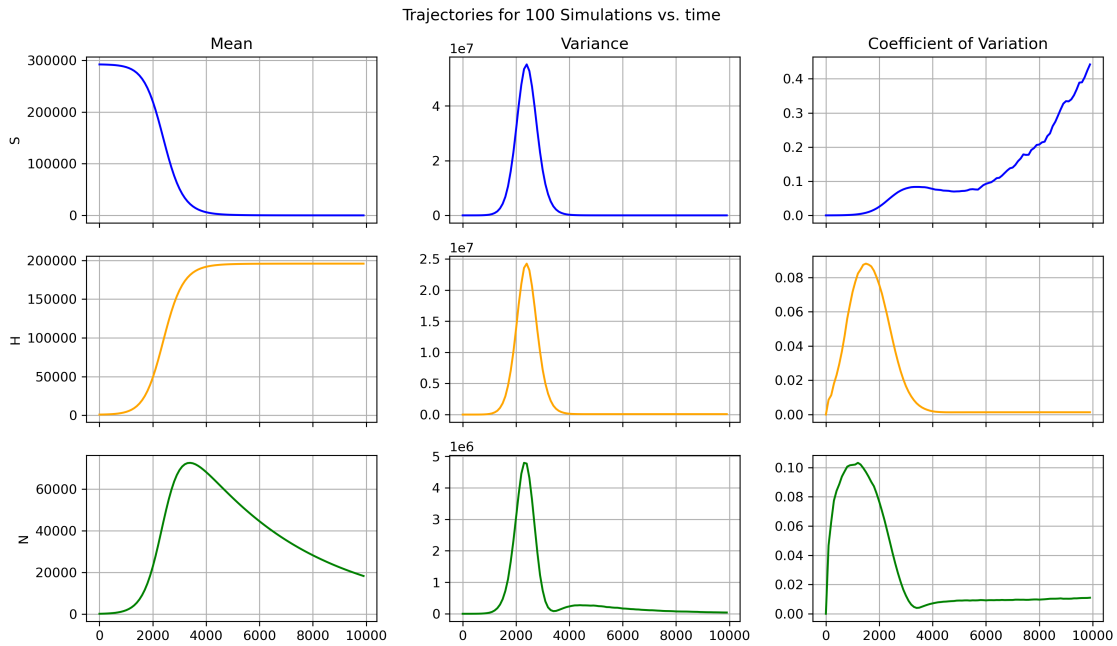


Figure 7: Trajectories of the 3 populations, split in mean, variance and variation coefficient over time for 100 simulations.